

MODERN APP DEVELOPMENT - 1

PROJECT REPORT

AUTHOR

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My name is Aryaman Kummamuru, and I am a second-year BS student at IIT Madras. I enjoy playing sports and have a strong interest in wildlife photography.

DESCRIPTION

The project I developed is a Quiz Master App. The goal of this project was to create a web application where users can attempt quizzes on topics of their choice and practice for exams. The app consists of two roles: an admin and regular users. The admin acts as a superuser, with the ability to manage subjects, chapters, quizzes, and more. The users are able to take quizzes and view their previous scores. The admin does not need to register manually, as I have built the registration process for the admin into the main.py file.

TECHNOLOGIES USED

1. **Flask-SQLAlchemy:** The primary technology used in this project is SQLAlchemy. It is an Object Relational Mapping (ORM) tool designed to define and manage database models. I used this tool to manage database operations such as reading, updating, and deleting records.
2. **Flask:** Flask serves as the web application framework for this project. It handles routing and the mapping of various pages. It also facilitates the management of form submissions like login and search operations.
3. **SQLite:** SQLite is the backend database used in this project. It is a lightweight, serverless database solution that allows me to store and manage data for users, subjects, and more.
4. **Jinja2:** Jinja2 is the default template engine in Flask. I used it to render dynamic content in the templates. For example, I employed Jinja2 in manage_users.html to iterate over the users in the database and display them on the page. I used it in other HTML pages as well to serve a similar purpose.
5. **Datetime:** The datetime module was used to store the date and time when a user attempts a quiz. This information is displayed in the table of previously attempted quizzes for the user's reference.

DB DESIGN

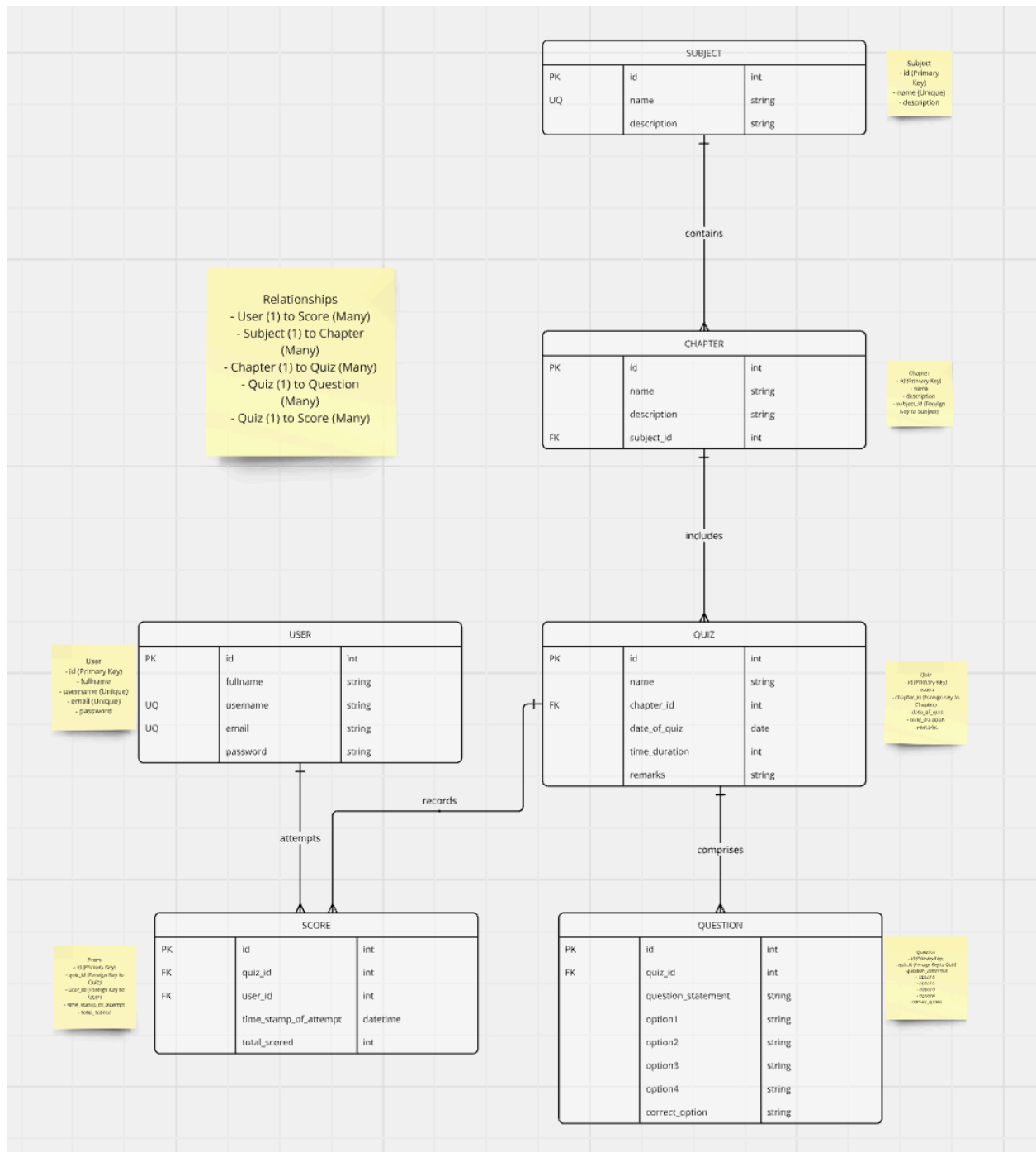
In accordance with the project requirements, I designed the database models with the following relationships:

1. One user can have many scores (One-to-Many).
2. One subject can have multiple chapters (One-to-Many).
3. One chapter can have multiple quizzes (One-to-Many).
4. One quiz can have multiple questions (One-to-Many).
5. One quiz can have multiple scores (One-to-Many).

I have attached an image of my Entity-Relationship Diagram (ERD) to illustrate the relationships between the classes. Please note the following abbreviations:

- **PK:** Primary Key

- **FK:** Foreign Key
- **UQ:** Unique



ARCHITECTURE AND FEATURES

I organized the project folder into different subfolders based on their contents and functionality.

- The **applications** folder contains the model.py, database.py and routes.py files.
- The **instance** folder contains the database that stores all the entered data.
- The **templates** folder contains the HTML files for the app, including home.html, login.html, register.html, admin_dashboard.html, etc.

- The **requirements.txt** file lists all the dependencies that I initially planned to use for the project.
- The **main.py** file is the core program that runs the web application and contains the code to create the admin without needing them to register manually.

Some of the key controllers I used include:

1. `@app.route('/')`
2. `@app.route('/login', methods = ['GET', 'POST'])`
3. `@app.route('/register', methods = ['GET', 'POST'])`
4. `@app.route('/admindashboard')`

VIDEO

Link:-  MAD1projectVid.mov